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**CSC3209 - Software Architecture and Design Pattern – Practical 7 Sample Solution**

**Answer the following questions:**

- 1- What are the uses of architecture documentation in your project?

**Sample Answer:**

**Three Uses for Architecture Documentation:**

**Education**

- Introducing people to the system
  - New members of the team
  - External analysts or evaluators
  - New architect

**Primary vehicle for communication among stakeholders**

- Especially architect to developers
- Especially architect to future architect!

**Basis for system analysis and construction**

- Architecture tells implementers what to implement.
- Each module has interfaces that must be provided and uses interfaces from other modules.
- Documentation can serve as a receptacle for registering and communicating unresolved issues.
- Architecture documentation serves as the basis for architecture evaluation

Refer to lecture notes for more detailed answer.

- 2- What are the notations to document an architecture? Which notation do you use in your documentation and why?

**Sample Answer:**

- *Informal notations*
  - Views are depicted (often graphically) using general-purpose diagramming and editing tools and visual conventions chosen for the system at hand.
  - The semantics of the description are characterized in natural language
  - They cannot be formally analyzed
  - The most common tool for informal notations is PowerPoint.

- *Semiformal notations*
  - Views are expressed in a standardized notation that prescribes graphical elements and rules of construction
  - Lacks a complete semantic treatment of the meaning of those elements
  - Rudimentary analysis can be applied to determine if a description satisfies syntactic properties.
  - UML is a semiformal notation in this sense.
- *Formal notations*
  - Views are described in a notation that has a precise (usually mathematically based) semantics.
  - Formal analysis of both syntax and semantics is possible.
  - Architecture description languages (ADLs)
  - Support automation through associated tools.

| Notation/Criteria | Preciseness of Specification | Analysis Support | Notation/Language Examples   | Tools                                | Industrial Domains Examples                                                                                                           |
|-------------------|------------------------------|------------------|------------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Informal          | Low                          | Low              | Informal diagrams, English   | PowerPoint, Informal Diagram Editors | System aspects that do not require specification of critical properties (e.g. foodpanda)                                              |
| Semi-formal       | Medium                       | Medium           | UML                          | UML Diagram Editors                  |                                                                                                                                       |
| Formal            | High                         | High             | B-Method, Event-B, Z Method, | Atelier-B [1], Rodin [2]             | Safety-critical systems (e.g. control system of autonomous Train: Paris_Métro_Line_14 that uses B-Method for formal verification [3]) |

Refer to lecture notes for a more detailed answer.

- 3- There are many views for architectural documentation. How do you decide which view to use in your documentation?

**Sample Answer:**

There are several views: Module Views, Component-and-Connector Views, and Allocation Views. Different views support different goals and uses. We do not advocate a particular view or collection of views. The views you should document depend on the uses you expect to make of the documentation. Each view has a cost and a benefit; you should ensure that the benefits of maintaining a view outweigh its costs.

Refer to lecture notes (Lecture10) for a more detailed answer on the need for every category of a view and the need for each type specifically.

- 4- Consider you are an architect in a company. Present the architectural document and give examples of every required section in the document.

**Sample Answer:** Refer to the attached file as a complete example of architecture document.